Name:
Shobhit Rai - 20251502032
Krish Kumar- 20251502015
Tamanna Sharma-20251502034
Siddhi Vyas-20251501163
Pranjali Nama-20251502022

QUESTION 1Duration Adder
A scheduling application needs to add two time durations together. Each duration is expressed in
hours, minutes, and seconds.
Write a C++ program that defines a class Duration to represent a time duration. The class should
store hours, minutes, and seconds as separate integer values.
The class should provide the following operations:
Accept a duration from the user.
Add two Duration objects together and return a new Duration object representing the sum. The addition must handle all necessary carry-over (e.g., 75 seconds should be converted to 1 minute and 15 seconds, 80 minutes should be converted to 1 hour and 20 minutes).
Display a duration in the format H hours, M minutes, S seconds.
In main(), create two Duration objects, accept input for both, compute their sum, and display all three durations (the two inputs and the result).

```cpp
#include <iostream>
using namespace std;

class Duration {
private:
   int hours;
   int minutes;
   int seconds;

public:
   Duration(int h = 0, int m = 0, int s = 0) {
       hours = h;
       minutes = m;
       seconds = s;
   }

   void input() {
       cin >> hours >> minutes >> seconds;
   }

   Duration operator+(const Duration &d) {
       Duration temp;
```

```
        temp.seconds = seconds + d.seconds;
        temp.minutes = minutes + d.minutes;
        temp.hours = hours + d.hours;

        temp.minutes += temp.seconds / 60;
        temp.seconds %= 60;


        temp.hours += temp.minutes / 60;
        temp.minutes %= 60;

        return temp;
    }
    void display() {
        cout << hours << " hours, "
             << minutes << " minutes, "
             << seconds << " seconds";
    }
};

int main() {
    Duration d1, d2, sum;

    cout << "Enter Duration 1 (hours minutes seconds): ";
    d1.input();

    cout << "Enter Duration 2 (hours minutes seconds): ";
    d2.input();

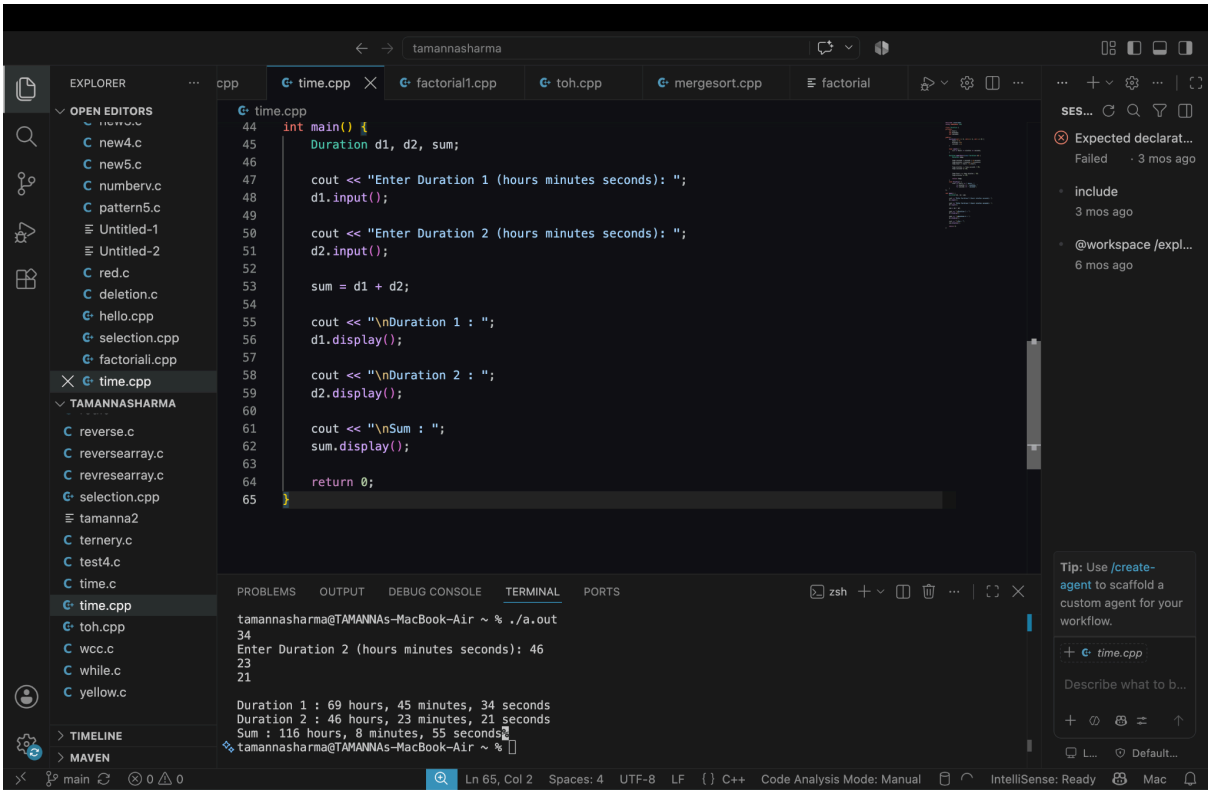
    sum = d1 + d2;

    cout << "\nDuration 1 : ";
    d1.display();

    cout << "\nDuration 2 : ";
    d2.display();

    cout << "\nSum : ";
    sum.display();

    return 0;
}
```



Section B — Array of Objects

Question 2 — Employee Bonus Report
A company's HR department needs a tool to compute annual bonuses for its employees based on
their years of experience, and to generate department-wise reports.
Write a C++ program that defines a class Employee with the following data members:
Employee ID (integer)

Name (string)
Department (string)
Base salary (float)
Years of experience (integer)
The class should provide the following operations:
Accept employee details from the user.
Calculate the annual bonus according to these rules:
More than 10 years of experience → 20% of base salary
More than 5 years (but ≤ 10) → 10% of base salary
5 years or fewer → 5% of base salary
Display the employee's complete record including the computed bonus amount.
In main(), perform the following:
1. Accept the number of employees N (assume N ≤ 100).
2. Create an array of N Employee objects and accept details for each.
3. Display the records of all employees in a tabular format, including each employee's bonus amount.

4. Identify and display the employee who receives the highest bonus amount.
5. Accept a department name from the user and display all employees belonging to that department. If no employee is found, display an appropriate message.
Notes:
You may assume all names and department names are single words.

```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;
class Employee
{
private: // Encapsulation: can be directly accessed from inside the employee class
    int id;
    string name;
    string deparment;
    float basesalary;
    int experience;
    float bonus;

public:
    void acceptDetails()
    {
        cin >> id >> name >> deparment >> basesalary >> experience;
        calcbonus();
    }
    void calcbonus()
    {
        if (experience > 10)
        {
            bonus = 0.20f * basesalary;
        }
        else if (experience > 5)
        {
            bonus = 0.10f * basesalary;
        }
        else
        {
            bonus = 0.05f * basesalary;
        }
    }
    void displayrecord()
    {
        cout << left << setw(10) << id << setw(15) << name << setw(15) << deparment <<
setw(15) << basesalary << setw(15) << experience << setw(15) << bonus << "\n";
    }
    float getBonus() const { return bonus; }
    string getDepartment() const { return deparment; }
    string getName() const { return name; }
};
int main()
{
    int n;
    cout << "No. of Employees: ";
```

```
    cin >> n;
    if (n > 100)
        n = 100;
    Employee emp[100];
    for (int i = 0; i < n; i++)
    {
        cout << "Enter details: " << i + 1 << " ";
        emp[i].acceptDetails();
    }
    cout << "EMPLOYEE REPORT: " << "\n";
    // cout << left << setw(10) << "ID" << setw(15) <<
    cout << left << setw(10) << "ID" << setw(15) << "Name" << setw(15) << "Deparment" <<
setw(15) << "Basesalary" << setw(15) << "Experience" << setw(15) << "Bonus" << "\n";
    cout << "\n";

    for (int i = 0; i < n; i++)
    {
        emp[i].displayrecord();
    }
    if (n > 0)
    {
        int maxIndex = 0;
        for (int i = 1; i < n; ++i)
        {
            if (emp[i].getBonus() > emp[maxIndex].getBonus())
            {
                maxIndex = i;
            }
        }
        cout << "\nEmployee with the highest bonus: " << emp[maxIndex].getName()
             << " (Bonus: " << emp[maxIndex].getBonus() << ")\n";
    }
    string searchDept;
    cout<<"Enter department name to search:";
    cin>>searchDept;
    bool found=false;
    cout<<"\nemployee in "<<searchDept<<" department:\n";

cout<<left<<setw(10)<<"ID"<<setw(15)<<"Name"<<setw(15)<<"Department"<<setw(15)<<"salary"<<s
etw(15)<<"Experience"<<setw(15)<<"Bonus"<<"\n";
    for(int i=0;i<n;i++){
        if(emp[i].getDepartment()==searchDept){
            emp[i].displayrecord();
            found=true;
        }

    }
    if(!found){
        cout<<"No employee found in this department. \n";
    }
    return 0;
}
```

Output:-

```
No. of Employees: 3
Enter details: 1 2015 KRISH IT 50000 22
Enter details: 2 2032 SHOBHIT IT 60000 6
Enter details: 3 2034 TAMMANA CSE 40000 4
EMPLOYEE REPORT:
ID        Name          Deparment     Basesalary     Experience     Bonus

2015      KRISH         IT            50000          22             10000
2032      SHOBHIT       IT            60000          6              6000
2034      TAMMANA       CSE           40000          4              2000

Employee with the highest bonus: KRISH (Bonus: 10000)
Enter department name to search:IT

employee inITdepartment:
ID        Name          Department    salary         Experience     Bonus
2015      KRISH         IT            50000          22             10000
2032      SHOBHIT       IT            60000          6              6000
```

Section C — Friend Functions

Question 3 — Box Volume Comparator

A warehouse manager needs to compare the storage volumes of two shipping boxes to decide which one to use for a particular shipment.

Write a C++ program that defines a class Box with private data members for length, width, and height (all floating-point values).

The class should provide: A function to accept dimensions from the user. A function to display the box dimensions. Define a friend function compareBoxes that takes two Box objects as parameters and performs the following: Calculates the volume of each box by directly accessing their private data members. Displays both volumes. Reports which box has a greater volume (or if they are equal). Displays the difference in volume between the two boxes. In main(), create two Box objects, accept dimensions for both, and call the friend function to compare them. Think about this: Why is a friend function a suitable choice here instead of a regular member function? Consider which object would call the member function, and how naturally it would access the other object's private data.

Code:

```cpp
#include<bits/stdc++.h>
using namespace std;

class Box{
    private:
    double length, width, height;

    public:
    void accept(){
        cin>>length>>width>>height;
    }
```

```
    void display(){
        cout<<"Length is:"<<length<<endl<<"Width
is:"<<width<<endl<<"Height is:"<<height;
    }

    friend void comparison(Box B1, Box B2);

};
void comparison(Box B1, Box B2){
    double vol1 = B1.length*B1.width*B1.height;
    double vol2 = B2.length*B2.width*B2.height;

    cout<<"Volume of box 1 is: "<<vol1<<" cubic units"<<endl;
    cout<<"Volume of box 2 is: "<<vol2<<" cubic units"<<endl;

    if(vol1>vol2){
        cout<<"Box 1 has greater volume."<<endl;
        cout<<"Difference is: "<<(vol1-vol2)<<" cubic units"<<endl;
    } else if(vol1<vol2){
        cout<<"Box 2 has greater volume."<<endl;
        cout<<"Difference is: "<<(vol2-vol1)<<" cubic units"<<endl;
    } else{
        cout<<" Both boxes have equal volume."<<endl;
    }
}

int main(){
    Box B1, B2;

    cout<<"Enter dimensions of box 1: ";
    B1.accept();
    B1.display();

    cout<<endl;

    cout<<"Enter dimensions of box 2: ";
    B2.accept();
    B2.display();

    cout<<endl;

    comparison(B1, B2);
```

```
    return 0;
}
```

Output:

